

Jackie (Jiaqi) Zou

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Technical Skills

Languages: Python, Java, C++, SQL, HTML, CSS, JavaScript, JSON **Machine Learning & Data:** scikit-learn, Pandas, PyTorch, TensorFlow (basic), angChain, LangGraph, RAG pipeline design, agent workflows

Frameworks & Libraries: FastAPI, Streamlit, TailwindCSS, React/React Native, Vue.js, Stripe, MERN, Node.js, Express.js, Next.js **Developer Tools:** Git, GitHub, Linux/Unix, command-line tooling

Projects

Developer Portfolio Website (HTML, TailwindCSS, JavaScript, Vercel) Aug 2025 – current update GitHub

- Designed and deployed a responsive developer portfolio showcasing **5+ software engineering projects**
- Implemented interactive UI features including dark-mode toggle, animated project cards, responsive grid layouts, and scroll-triggered animations, Built modular front-end architecture separating **HTML structure, TailwindCSS styling, and JavaScript navigation logic**
- Organized project content and visual dashboards enabling recruiters to understand technical abilities **up to 2× faster**
- Deployed publicly using **GitHub Pages** with version-controlled updates

Backend Inventory Management System (Java, SQL) May 2025 – Aug 2025 GitHub

- Designed and implemented a backend inventory platform using **Java service-layer architecture**
- Built **RESTful APIs** supporting inventory tracking, order processing, and user management
- Integrated relational database schemas ensuring **transaction-safe operations and consistent data updates**
- Automated inventory workflows improving operational efficiency by **50% compared to manual tracking**
- Engineered backend logic to correctly handle **edge cases and concurrent request scenarios**

Consulting Platform — Business Intelligence System (Python, FastAPI, Streamlit, LLM) Sep 2025 – Jan 2026 GitHub

- Developed an end-to-end AI analytics platform transforming raw business datasets into **automated strategic recommendations**
- Built data pipelines performing **data cleaning, feature extraction, and machine learning analysis**
- Integrated **LLM-powered insight generation** translating analytical results into business strategy suggestions
- Developed an interactive **Streamlit dashboard** enabling non-technical users to visualize insights
- In simulated business testing scenarios, generated recommendations associated with **up to 2× potential sales improvement**

Stock Market Outperformance Prediction System (Python, Pandas, scikit-learn) Oct 2025 – Feb 2026 GitHub

- Built a machine learning pipeline predicting stocks likely to outperform the **S&P500**
- Developed preprocessing pipelines cleaning and aligning **large-scale financial time-series datasets**
- Engineered financial indicators by parsing Yahoo Finance statistics using **regex-based feature extraction**
- Trained a **Random Forest classifier** and evaluated performance through historical backtesting
- Achieved approximately **30% prediction precision** when identifying market outperformance signals

Education & Awards

University of Waterloo, Waterloo, Ontario Sep 2025 – Present

Honours Computer Science (Co-op) (Relevant Coursework: Designing Functional Programs, Elementary Algorithm Design and Data Abstraction, Tools and Techniques for Software Development, Object-Oriented Software Development, Linear Algebra, Cognitive Science, Statistics)

Canadian Open Mathematics Challenge (COMC) 2024: Honorable Mention; Best in Grade 12 (Central Toronto), Score: 66

Recognizes strong mathematical reasoning, proof skills, and advanced problem solving

Euclid Mathematics Contest 2025: School Champion; Honor Roll, Score: 91/100 (Top 100 nationally)

University-level mathematics competition emphasizing algorithmic thinking